

WEEK8-PRACTICAL

Task1:

Develop a program using `malloc()`, `calloc()`, `realloc()`, and `free()`. Monitor memory allocation behavior and identify memory leaks using Valgrind.

```
UW PIC0 5.09 File: dynamic.c Modified
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int i;

    printf("Using malloc()\n");

    int *a = (int *)malloc(3 * sizeof(int));

    for (i = 0; i < 3; i++)
    {
        a[i] = i + 1;
        printf("a[%d] = %d\n", i, a[i]);
    }

    printf("\nUsing calloc()\n");

    int *b = (int *)calloc(3, sizeof(int));

    for (i = 0; i < 3; i++)
    {
        printf("b[%d] = %d\n", i, b[i]);
    }

    printf("\nUsing realloc()\n");

    a = (int *)realloc(a, 5 * sizeof(int));

    a[3] = 4;
    a[4] = 5;

    for (i = 0; i < 5; i++)
    {
        printf("a[%d] = %d\n", i, a[i]);
    }

    printf("\nFreeing memory\n");

    free(a);
    free(b);

    printf("Memory freed successfully\n");

    return 0;
}
```

Get Help
Exit

WriteOut
Justify

Read File
Where is

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```
Last login: Thu Sep 17 13:41:49 on ttys001
(eval):8: command not found: >
vishalkarnati@VISHALS-MacBook-Air ~ % cd ~/Desktop/os
vishalkarnati@VISHALS-MacBook-Air os % mkdir week8spract
vishalkarnati@VISHALS-MacBook-Air os % cd week8spract
vishalkarnati@VISHALS-MacBook-Air week8spract % nano dynamic.c
vishalkarnati@VISHALS-MacBook-Air week8spract % gcc dynamic.c -o dynamic
vishalkarnati@VISHALS-MacBook-Air week8spract % ./dynamic
Using malloc()
a[0] = 1
a[1] = 2
a[2] = 3

Using calloc()
b[0] = 0
b[1] = 0
b[2] = 0

Using realloc()
a[0] = 1
a[1] = 2
a[2] = 3
a[3] = 4
a[4] = 5

Freeing memory
Memory freed successfully
vishalkarnati@VISHALS-MacBook-Air week8spract %
```

Task2:

Develop a program that demonstrates Copy-on-Write behavior after a `fork()`. Analyze memory usage before and after modifying data in the child process.

```
UW PICO 5.09 File: cow.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

#define SIZE 1000000

int main()
{
    int *data = malloc(SIZE * sizeof(int));

    for (int i = 0; i < SIZE; i++)
    {
        data[i] = i;
    }

    printf("Before fork()\n");
    printf("Parent PID: %d\n", getpid());
    printf("Address of data: %p\n", (void *)data);

    pid_t pid = fork();

    if (pid < 0)
    {
        printf("fork failed\n");
        free(data);
        return 1;
    }

    if (pid == 0)
    {
        printf("\nChild process\n");
        printf("Child PID: %d\n", getpid());
        printf("Address of data: %p\n", (void *)data);

        printf("Modifying data in child...\n");
        data[0] = 9999;
        printf("Child data[0] = %d\n", data[0]);

        sleep(10);
    }
    else
    {
        printf("\nParent process\n");
        printf("Parent PID: %d\n", getpid());
        sleep(5);
        printf("Parent data[0] = %d\n", data[0]);

        wait(NULL);
    }

    free(data);

    return 0;
}

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⌘ Exit ⌘ Justify ⌘ Where is ⌘ Next Pg ⌘ UnCut Text ⌘ To Spell
```

```
vishalkarnati@VISHALS-MacBook-Air week8spract % nano cow.c
vishalkarnati@VISHALS-MacBook-Air week8spract % gcc cow.c -o cow
vishalkarnati@VISHALS-MacBook-Air week8spract % ./cow
Before fork()
Parent PID: 2556
Address of data: 0x60a000000

Parent process
Parent PID: 2556

Child process
Child PID: 2557
Address of data: 0x60a000000
Modifying data in child...
Child data[0] = 9999
Parent data[0] = 0
vishalkarnati@VISHALS-MacBook-Air week8spract %
```